

Alberto Ruiz-Biestro

Physics Graduate Student

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EDUCATION

- Present **Ph.D. Physics & Astronomy**, *Rice University*
- 2020 – 2024 **B.Sc. Engineering Physics**, *Monterrey Institute of Technology*.
GPA: 96.68/100
TOEFL iBT Score: 108.

PUBLICATIONS

1. **Alberto Ruiz-Biestro** and Julio C. Gutierrez-Vega.
“Solutions of the Lippmann-Schwinger equation for confocal parabolic billiards”.
Phys. Rev. E., Mar 2024. [doi:10.1103/PhysRevE.109.034203](https://doi.org/10.1103/PhysRevE.109.034203).

CONFERENCE PRESENTATIONS

2. **Mexican Optics and Photonics Meeting**. Poster presentation.
“Lippmann-Schwinger equation in parabolic geometries”. Nov 2023.
1. **National Space Activity Congress** (CONACES). “Raman spectrometer design for biosignature detection” (virtual). Nov 2021.

AWARDS

- Apr 2023 **Best Team Project**. *International Centre for Theoretical Physics & Quantum Trieste*, Italy. 2nd place in the quantum hackathon.
Project: Krylov subspace methods in a Quantum Computer
- Aug 2020 **Academic Merit Scholarship**, *Monterrey Institute of Technology*.

NUMERICAL SKILLS

- Numerical* Proficient in **Julia**, **MATLAB**, **Python**, and **Linux**. Proven skills in **COMSOL** Wave Optics module, **Mathematica**, and **Git**.
- Software* Developed an EM 2D **scattering toolbox** based on boundary-integral method. Documentation available at <https://biestro.github.io/posts/boundary-wall/>

TEACHING EXPERIENCE

- Jan 2025 – Jun 2025 **TA for PHYS 202 Modern Physics** (Rice University).
Graded homework and exams.
- Aug 2023 – Dec 2023 **TA for Mathematical Methods** (Tec de Monterrey).
Graded homework and exams; held weekly advisory sessions.
- Aug 2022 – Jun 2023 **TA for Modern E&M** (Tec de Monterrey).
Graded homework and exams; held weekly advisory sessions.

LEADERSHIP

- 2022 – 2023 **Quantum Computing Club** co-founder and VP.
- Organization of seminars, including one with Dr. Benjamín Perez-García on the implementation of Deutsch's algorithm with linear optics.
 - Organization and construction of a variety of courses that gave undergraduate students tools to program and analyze quantum algorithms.
 - Active participation in the organization of my institution's first **quantum hackathon**. Helped with dissemination and spreading the invitation to external faculty and students.
 - Coordinated and teaching of workshops in collaboration with the *Physics Student Society* (AEF in Spanish) from Nuevo-Leon's Autonomous University (UANL).
 - Organization, planning, and direction of quantum computing bootcamps, offering intensive courses to students from ITESM as well as from other universities.
 - Our outreach has grown beyond the state of Nuevo León.
- 2023 SPIE Student Chapter President
- 2022 – 2023 Given talks and short courses on Julia, Python, and $\text{\LaTeX}$.

RESEARCH EXPERIENCE

- Sep 2023 – Present **Photonics and Mathematical Optics Group, Monterrey, Mexico**
Advisors: Julio C. Gutierrez-Vega 
Implemented a Boundary Integral Method for solving the Lippmann-Schwinger (scattering) Equation.
Development of meshes for discretization and parallel computation.
Advanced theoretical methods and mathematical formulations for analytic results.
- Apr 2023 **International Centre for Theoretical Physics & Quantinuum, Trieste, Italy**
Advisor: Nathan Fitzpatrick  (*Quantinuum*)
- Generated ground and excited state curves using a Quantum Krylov-subspace method along a reaction coordinate for an H_2 molecular Hamiltonian.
 - Development of hybrid quantum-classical algorithms with TKET and the InQuanto quantum chemistry platform; aided team in setting up and using **Git** for version control.
 - Collaborated with graduate students from diverse backgrounds. Our team received the *Best Team Project* award, along with second place.
- Aug 2021 – Jun 2022 **Photonics and Mathematical Optics Group, Monterrey, Mexico**
Advisors: Dr. Antonio Ortiz-Ambriz  Dr. Gerardo Fox  Dr. Servando López 
- Numerical simulation of the *Nonlinear Schrodinger Equation* through *pseudo-spectral method* (split-step Fourier) and numerical solutions of Boundary Value Problems (shooting method, finite differences, etc.).
 - Developed audio-identification algorithm in order to identify an audio recording from a microphone (FFT and signal-processing methods).
 - Analyzed the travelling-salesman-problem through simulated annealing; simulated the dynamics and critical points of the Lenz-Ising model.

- Developed **Genetic algorithms** and **Neural Networks**; Experience with **Agent Based Modeling**.